Questions by topic: The atmosphere: energy processes (UV, IR, visible light) - Set 1 (60 marks)

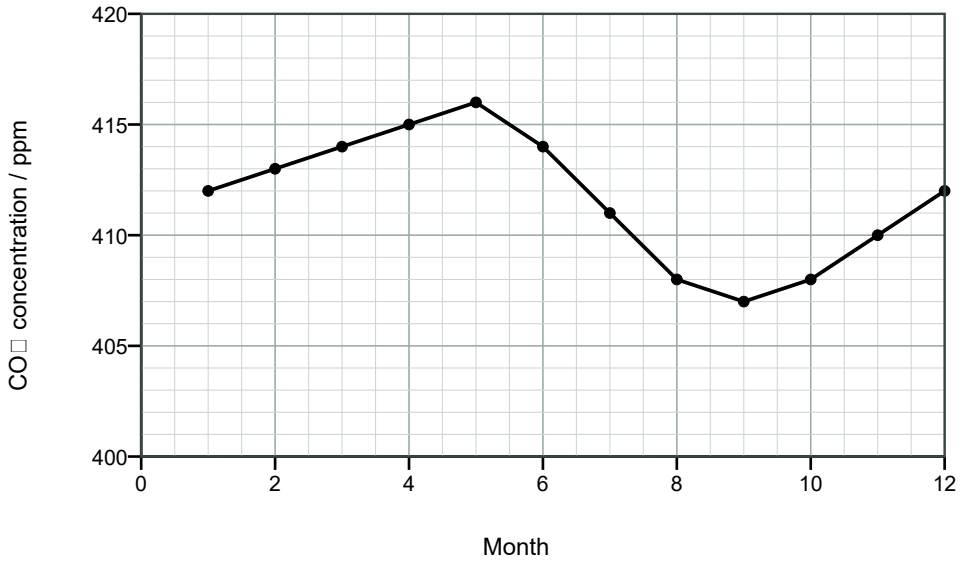
Answer all questions in the spaces provided.

0 1 Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1



0 1 . 1 Describe and explain the pattern shown in Figure 1.

[4 marks]

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0 1 . 2 Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

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0 1 . 3 Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

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0 1 . 4 Name the two layers of the atmosphere most affected by human activities.

[2 marks]

1 .....

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0 2 Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

0 2 . 1 Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer ____ %

0 2 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

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Ultraviolet (UV) .....

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Visible light .....

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Infrared (IR) .....

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0 2 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

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0 2 . 4 Explain the term **thermal stratification** of the atmosphere.

[1 mark]

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0 3 . 1 Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

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0 3 . 2 Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

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0 3 . 3 Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

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0 3 . 4 Give **two** industrially important gases extracted from the atmosphere.

[2 marks]

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0 4 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 4 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 4 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 5 Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 5 . 1 Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

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Answer [ ] °C

0 5 . 2 Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

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0 5 . 3 Explain what is meant by a **greenhouse gas**.

[1 mark]

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0 5 . 4 The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

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0 5 . 5 Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

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0 6 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

0 6 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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0 6 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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0 6 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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0 6 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Extra space

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END OF QUESTIONS